

DevTracker

Internal Development Management System

Project Objective:-

DevTracker is an enterprise-grade developer activity and productivity management platform developed for H.N. Reliance Hospital to centralize task tracking, work logging, and performance monitoring across development teams.

The system enables developers and interns to record daily work, manage tasks, track work sessions using an integrated timer, and submit leave requests, while providing administrators and managers with real-time dashboards, productivity analytics based on a 40-hour work week, and comprehensive reporting.

The platform supports secure Reliance SSO/JWT authentication, role-based access control, Excel-based bulk data import/export, notifications, upload history, audit monitoring, and centralized team management. By automating activity tracking and productivity analysis, DevTracker improves project visibility, streamlines reporting, enhances workforce planning, and enables data-driven decision-making across the organization.



1. Project Overview & Technology Stack

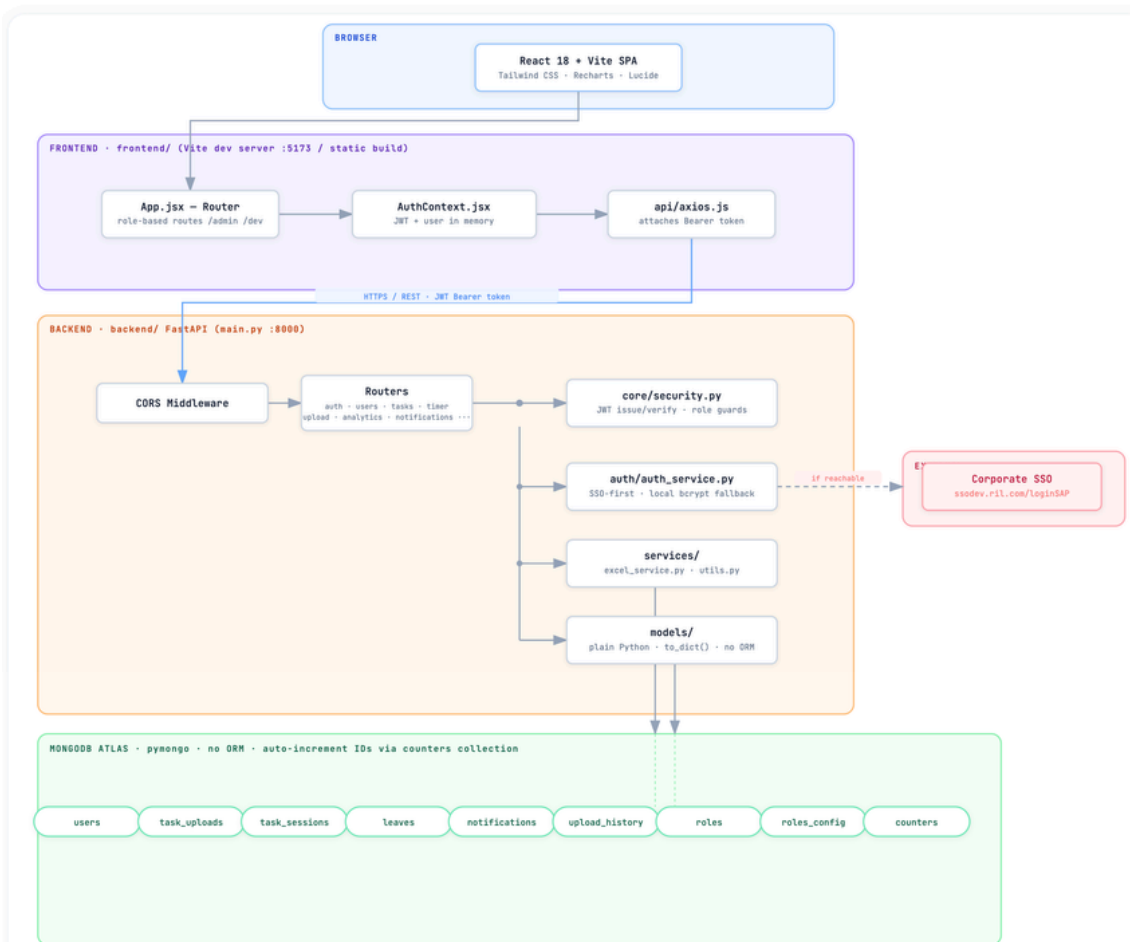
Layer	Technology
Frontend	React + Vite + Tailwind CSS + Axios + Recharts
Backend	Python FastAPI (REST APIs)
Database	MongoDB
ODM	Motor (Async MongoDB Driver)
Authentication	JWT Token Based Authentication (SSO Integration with Reliance SSO)
File Storage	Azure Blob Storage (for Excel Uploads)
Excel Processing	Pandas + OpenPyXL
Background Tasks	Celery + Redis (for async tasks & notifications)
Deployment	Microsoft Azure (App Service + Azure Database for MongoDB)
Monitoring & Logging	Azure Monitor + Application Insights

Core Modules: Secure Login & Reliance SSO Authentication, Role-Based Access Control, Admin Dashboard & Analytics, Developer Dashboard with Task & Time Tracking, Weekly Excel Upload & Validation, Productivity Calculator (40-Hour Work Week), Leave Management, Work Session Tracking, Task Assignment & Monitoring, Notifications, Reports & Export, Upload History, Audit Trail & Activity Monitoring, and Domain-Based User Validation.

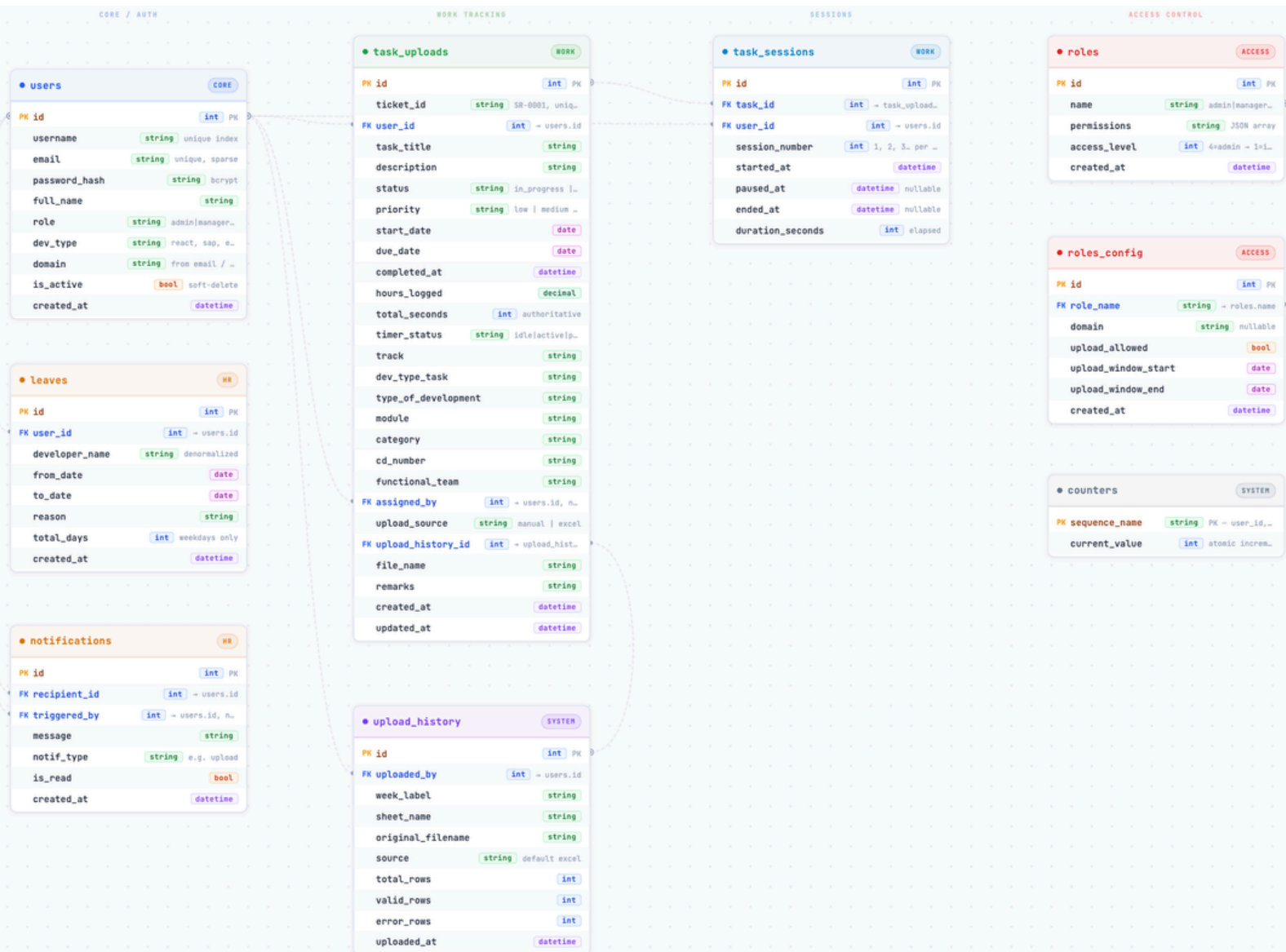
2. System Architecture & Database Design

System Architecture

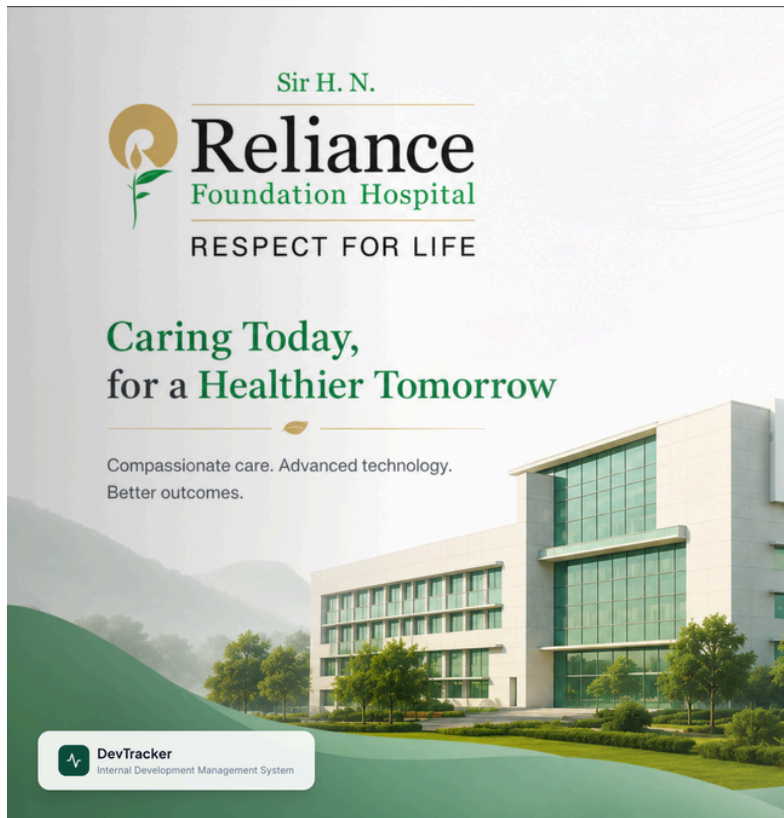
React 18 • FastAPI • MongoDB Atlas • JWT Auth • SSO



The application follows a modern three-tier architecture, with React powering the frontend, FastAPI serving secure REST APIs and business logic, and MongoDB providing a flexible document-oriented database for managing users, task uploads, work sessions, notifications, leave records, upload history, and role-based access control. The backend enforces authentication and role-based authorization, while Azure hosts the production environment for secure, scalable, and reliable deployment.



3. Developer Portal



Welcome Back

Sign in with your Reliance credentials

USERNAME

your.username

PASSWORD

Enter your password

☐ Keep me signed in

[Forgot Password?](#)

Sign In

Need help? [Contact IT Support](#)

v2.4.1 - build 20240619

DevTracker
H.N. Reliance Hospital

MENU

- My Dashboard
- My Logs
- Add Log
- Upload Excel
- My Reports

Arjun Intern
INTERN

My Dashboard

Wednesday, 8 July 2026

Good evening, Arjun 🌞
No active task
Start a task to begin tracking time

TODAY
4h 0m
Logged today

ACTIVE TASK
—
None running

WIP TASKS
0
0 paused

DONE TODAY
1
This week: 0.3h

My Tasks

Only one task can run at a time — switching auto-pauses the current one

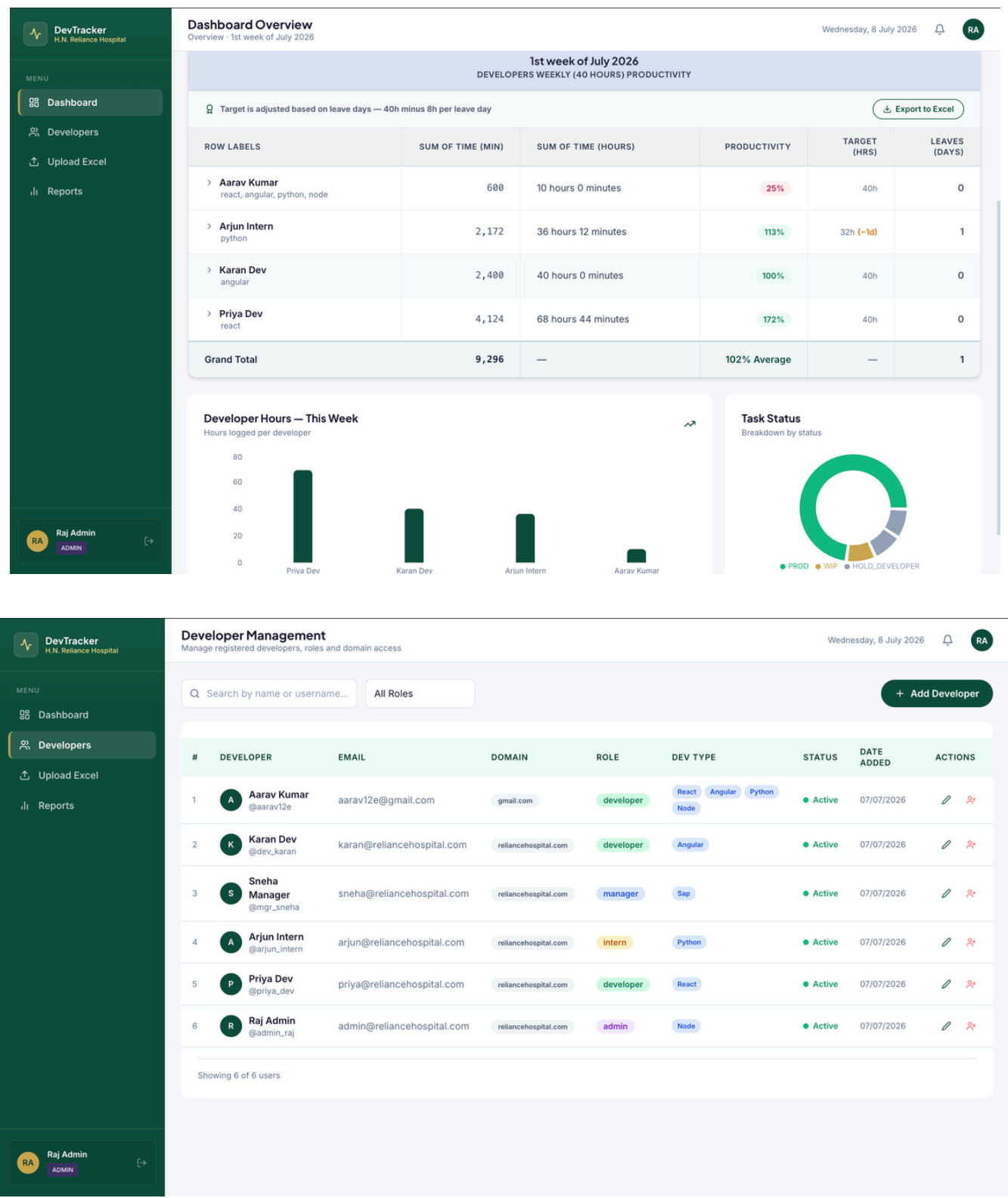
+ Start New Task

No active tasks. Start a task to begin the timer.

PENDING TASKS

- Idle SR-0013 RFH Medium
this is test 12
Production Issue - rajul
00:00:00
Start
- Idle SR-0005 RFH High Priority
this s test 5 edited status
00:00:00
Start

The Admin Portal provides complete visibility into development activities across teams. Administrators can monitor logs, review developer performance, upload weekly reports, manage users, and generate exports for management reporting.



4. Reports Export & Import

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H.N. Reliance Hospital

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Arjun Intern
AI
INTERN

New Development Log

Fill all required fields accurately

Wednesday, 8 July 2026

AI

Log Development Task

Request Leave

New Development Log Entry

Clear Form

X Cancel

Submit Log

A serial number (SR-XXXX) will be auto-assigned when you save this log.

TRACK

DEV TYPE

TYPE OF DEVELOPMENT

REFERENCE NUMBER

RFH

Select Dev Type

Select Type

CD Number e.g. 8089020

DEVELOPMENT SUBJECT

MODULE

CATEGORY

e.g. BPL Case Discount SOA and Invoice Print

Select Module

Select Category

FUNCTIONAL TEAM

ASSIGNEE (DEVELOPER/INTERN)

START DATE

END DATE

e.g. Biswajit

Arjun Intern You

dd/mm/yyyy

dd/mm/yyyy

PRIORITY

STATUS

TIME LOGGED (MINS)

Low Medium High

WIP

0

DEVELOPMENT DESCRIPTION

REMARKS

Detailed explanation of the development...

Additional notes or context...

All fields are optional. Leave empty to default.

DevTracker
H.N. Reliance Hospital

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Arjun Intern
AI
INTERN

Bulk Upload

Upload weekly developer status sheet (.xlsx format only)

Wednesday, 8 July 2026

AI

Format Requirements

Download Template

Ensure your Excel file follows the standard template. Required columns: Development Subject. Optional: Track, Dev Type, Type of Development, CD, Functional Team, Developers, Start Date, End Date, Time (Min), Status, Remarks.

Drag & drop your .xlsx file here

or browse file

Max file size: 10MB · .xlsx and .xls only

Upload History

Load History

Click "Load History" to see past uploads

DevTracker uses Reliance Single Sign-On (SSO) for secure user authentication and centralized access management. The system implements role-based access control (Admin, Manager, Developer, and Intern), secure REST APIs, session tracking, task management, notifications, and audit logging to ensure data security and operational transparency. Built using React, FastAPI, and MongoDB, the application is deployed on Microsoft Azure with secure HTTPS communication and scalable cloud infrastructure. This enables centralized task reporting, real-time work tracking, improved visibility, and efficient development management across teams.